

Tutorial 8: R Programming, Quantile and Value at Risk (STUDENT VERSION)

Theme: From Programming Loops to Quantiles, VaR, cVaR, Portfolio Risk and Tail-Risk Thinking

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Programme: Nurturing Future Risk Managers and Actuaries: An AI-powered Journey for Gifted
Secondary Students

Synergy & Comparison

The Logic of this Tutorial: Basic principles of R programming, functions, loops, distribution functions, quantiles, Value-at-Risk and conditional Value-at-Risk form our foundation. Tutorial 7 strengthened students' understanding of distributions, simulation, empirical CDFs, quantiles, Poisson approximation and the Central Limit Theorem. Tutorial 8 brings these ideas together into a coherent risk-management workflow. Students will use R to move from programming structure to simulated data, returns, losses, quantiles, VaR, cVaR, portfolio risk and visual interpretation.

Key Definition / Result

Learning Objectives. By the end of this tutorial, students should be able to:

1. use functions, if statements and for loops in simple risk-modelling tasks;
2. explain quantiles as cut-off values of a distribution or dataset;
3. select and use the correct theoretical quantile function in R (`qnorm`, `qunif`, `qpois`, `qbinom`);
4. distinguish between lower-tail return risk and upper-tail loss risk;
5. calculate empirical quantiles using `quantile()` in R;
6. define and interpret Value-at-Risk (VaR);
7. define and interpret conditional Value-at-Risk (cVaR);
8. compare empirical, parametric Normal and Monte Carlo approaches;
9. explain how portfolio weights, diversification and correlation affect portfolio VaR;
10. write simple R functions for VaR and cVaR;
11. use graphs to visualise tail risk;

12. recognise that a risk number is meaningful only when we understand the assumptions behind it.

A. Programming and Loops: Building Blocks for Risk Calculation

Synergy & Comparison

Why begin with programming? Students will use R functions, if statements and for loops to write simple reusable calculations. In Tutorial 7, fundamental programming concepts were connected to simulation and statistical distributions. In this tutorial, we apply these programming tools in a purposeful way: they are not isolated coding tricks, but the building blocks that help us calculate quantiles, Value-at-Risk and conditional Value-at-Risk in a clean and repeatable manner.

Key Definition / Result

In R, three programming structures are especially important for risk modelling:

Structure	Syntax	Purpose in risk modelling
Function	<code>function(arg1, ...){ expr }</code>	Define a reusable calculation
If statement	<code>if(cond){ expr }</code>	Make a decision when a condition is true
For loop	<code>for(name in expr1){ expr2 }</code>	Repeat a calculation or simulation many times

Suggested AI Prompt: Programming and Loops for Risk Measurement ★ Core

“Explain what functions, vectorised conditions, and for loops are in R. Show simple examples of how they can be used in risk management, especially for classifying losses and repeating simulations. Please use clear variable names and beginner-friendly comments.”

Worked Example: An Indicator Function for Tail Events ★★ Intermediate

An indicator function is very useful in probability and risk measurement. For exam-

ple, we may want to count whether a loss lies between two values a and b :

$$I_{[a,b]}(x) = \begin{cases} 1, & a \leq x \leq b, \\ 0, & \text{otherwise.} \end{cases}$$

Vectorised version in R:

```
indicator <- function(x, a, b) {
  (a <= x) * (x <= b)
}

loss_values <- c(10, 25, 80, 120, 200)
indicator(loss_values, 20, 100)
sum(indicator(loss_values, 20, 100))
```

Investigation / Activity: From Indicator Functions to Tail Losses

★★

Intermediate

Task. Suppose we have the following loss values:

10, 25, 80, 120, 200, 300.

Use R to check which losses are at least 100, then calculate the average of those tail losses.

R Code:

```
loss_values <- c(10, 25, 80, 120, 200, 300)
tail_indicator <- loss_values >= 100
tail_indicator
tail_losses <- loss_values[tail_indicator]
tail_losses
mean(tail_losses)
```

Investigation / Activity: Repeating Simulations with a For Loop

★★

Intermediate

Task. Suppose the number of claims in a day follows a Poisson distribution. Each claim is assumed to cost 1000 monetary units. Repeat the simulation 10 times. Each time, generate 1000 daily losses and calculate the 95% quantile.

R Code:

```
set.seed(2026)
number_of_repetitions <- 10
```

```

sample_size <- 1000
VaR_results <- rep(0, number_of_repetitions)

for (repeat_index in 1:number_of_repetitions) {
  claim_counts <- rpois(sample_size, lambda = 4)
  simulated_losses <- 1000 * claim_counts
  VaR_results[repeat_index] <- quantile(simulated_losses, 0.95)
}

VaR_results
mean(VaR_results)
sd(VaR_results)

```

Reflection Question: Simulation Variability

If each simulation uses the same Poisson model but generates a new random sample, why might the estimated 95% VaR change from one repetition to another?

B. Quantiles: The Language of Cut-Off Values

Key Definition / Result

For a random variable X , the p -quantile is a value q_p such that

$$P(X \leq q_p) = p.$$

For data, the sample p -quantile is an empirical cut-off value estimated from observations. In R, this is calculated as:

```
quantile(data, p)
```

For **theoretical distributions**, R provides specific functions starting with **q** to find the exact cut-off value without needing simulated data. You must choose the function that matches your risk model assumption:

- **Normal** (Continuous, bell-shaped, e.g., stock returns): `qnorm(p, mean, sd)`
- **Uniform** (Continuous, flat, equally likely bounds): `qunif(p, min, max)`
- **Poisson** (Discrete, counting events like claims): `qpois(p, lambda)`
- **Binomial / Bernoulli** (Discrete, successes in trials): `qbinom(p, size, prob)`

Worked Example: Choosing the Right Theoretical Quantile Function ★

Core

Question. How do we find the 95% theoretical cut-off (quantile) for different types of risk models?

R Code:

```
# 1. Normal: Asset returns assumed to be bell-shaped
# Mean = 0, Standard Deviation = 1
qnorm(0.95, mean = 0, sd = 1)

# 2. Uniform: All outcomes between 0 and 100 are equally likely
# Min = 0, Max = 100
qunif(0.95, min = 0, max = 100)

# 3. Poisson: Counting car accidents per day
# Average rate = 4 accidents per day
qpois(0.95, lambda = 4)

# 4. Binomial: Number of defaults in 100 loans
# Probability of default = 5% (0.05)
qbinom(0.95, size = 100, prob = 0.05)
```

Worked Example: Sample Quantiles from a Small Dataset ★ Core

Question. Consider the following values:

2, 5, 7, 9, 12, 15, 20, 30.

Calculate the 25%, 50%, 75% and 95% sample quantiles.

R Code:

```
values <- c(2, 5, 7, 9, 12, 15, 20, 30)
quantile(values, c(0.25, 0.50, 0.75, 0.95))
```

Investigation / Activity: Quantiles from Poisson Claim Losses ★★

Intermediate

Task. Simulate 1000 daily claim counts from a Poisson distribution. Suppose each claim costs 1000 monetary units. Then calculate the 90%, 95% and 99% quantiles of daily loss.

R Code:

```
set.seed(2026)
```

```

claim_counts <- rpois(1000, lambda = 4)
losses <- 1000 * claim_counts
quantile(losses, c(0.90, 0.95, 0.99))

hist(
  losses,
  breaks = 20,
  main = "Histogram of Simulated Poisson Claim Losses",
  xlab = "Daily Loss",
  col = "lightblue",
  border = "white"
)

```

Investigation / Activity: Adding a Bernoulli Shock Event ** Intermediate

Task. Suppose that, on most days, losses come from ordinary claim counts. However, with probability 5%, a shock event adds an extra loss of 8000 monetary units. Use a Bernoulli random variable to represent this shock.

R Code:

```

set.seed(2026)
sample_size <- 1000
ordinary_claim_counts <- rpois(sample_size, lambda = 4)
ordinary_losses <- 1000 * ordinary_claim_counts

shock_indicator <- rbinom(sample_size, size = 1, prob = 0.05)
shock_losses <- 8000 * shock_indicator

total_losses <- ordinary_losses + shock_losses
quantile(total_losses, c(0.90, 0.95, 0.99))

hist(
  total_losses,
  breaks = 30,
  main = "Losses with Rare Bernoulli Shock Events",
  xlab = "Total Daily Loss",
  col = "lightblue",
  border = "white"
)

```

Common Mistake

Do not say that a 95% quantile is a value that occurs with probability 95%. It is a

cut-off value. Saying $P(X \leq q_{0.95}) = 0.95$ means 95% of the distribution lies at or below the quantile.

C. Returns, Losses and Direction of Risk

Synergy & Comparison

Important convention: In finance and risk management, we must be careful about direction. If we study **returns**, bad outcomes are often in the lower tail. If we study **losses**, bad outcomes are often in the upper tail. VaR can be expressed using either convention, but the interpretation must be clear.

Key Definition / Result

Suppose R is a return and L is a loss. A simple relationship is:

$$L = -R.$$

If a return is very negative, the corresponding loss is very positive.

For returns, risk often concerns lower-tail quantiles such as:

$$q_{0.05}(R).$$

For losses, risk often concerns upper-tail quantiles such as:

$$q_{0.95}(L).$$

Worked Example: Converting Returns to Losses

★ Core

Question. Suppose we observe daily returns. Convert them to losses and compare their quantiles.

R Code:

```
returns <- c(0.01, -0.02, 0.005, -0.04, 0.03, -0.01, -0.06, 0.02)
losses_from_returns <- -returns

quantile(returns, c(0.05, 0.25, 0.50))
quantile(losses_from_returns, c(0.50, 0.75, 0.95))
```

Reflection Question: Direction of Risk

If a company reports a very low return on a particular day, why might the corresponding loss be very high?

D. Value-at-Risk: A Quantile-Based Risk Measure**Key Definition / Result**

Value-at-Risk for a loss variable. If L is a loss, the VaR at confidence level α is commonly defined as:

$$\text{VaR}_\alpha(L) = q_\alpha(L).$$

This means:

$$P(L \leq \text{VaR}_\alpha) = \alpha.$$

For example, the 95% VaR is a loss cut-off such that about 95% of losses are below it and about 5% of losses exceed it.

Worked Example: Empirical VaR from Simulated Poisson Losses

★★

Intermediate

Question. Simulate 1000 daily losses using Poisson claim counts and calculate the 95% empirical VaR.

R Code:

```
set.seed(2026)
claim_counts <- rpois(1000, lambda = 4)
losses <- 1000 * claim_counts
VaR_95 <- quantile(losses, 0.95)
VaR_95
```

Investigation / Activity: Writing a VaR Function

★★ Intermediate

Task. Write a reusable function that calculates VaR from a vector of losses and a confidence level.

R Code:

```
calculate_VaR <- function(loss_values, alpha) {
  quantile(loss_values, alpha)
}
```



```
calculate_VaR(losses, 0.90)
calculate_VaR(losses, 0.95)
calculate_VaR(losses, 0.99)
```

Investigation / Activity: Visualising VaR on a Histogram ** Intermediate

Task. Draw a histogram of simulated losses and mark the 95% VaR.

R Code:

```
hist(
  losses,
  breaks = 20,
  main = "Simulated Poisson Losses with 95% VaR",
  xlab = "Daily Loss",
  col = "lightblue",
  border = "white"
)
abline(v = VaR_95, col = "red", lwd = 3, lty = 2)
legend(
  "topright",
  legend = "95% VaR",
  col = "red",
  lwd = 3,
  lty = 2,
  bty = "n"
)
```

Common Mistake

VaR does not tell us how bad the loss can be after the VaR threshold is exceeded. It tells us where the tail begins, not the average size of losses inside the tail.

E. Conditional Value-at-Risk: Looking Beyond VaR

Key Definition / Result

Conditional Value-at-Risk. For a loss variable L , the conditional Value-at-Risk at level α may be interpreted as the average loss given that the loss is beyond the VaR threshold:

$$\text{cVaR}_\alpha(L) = E[L \mid L \geq \text{VaR}_\alpha(L)].$$

It is also commonly called expected shortfall. It gives information about the severity of losses in the tail.

Worked Example: Empirical cVaR from Simulated Poisson Losses

★★

Intermediate

Question. Using the same simulated losses, calculate the 95% cVaR.

R Code:

```
VaR_95 <- quantile(losses, 0.95)
tail_losses <- losses[losses >= VaR_95]
cVaR_95 <- mean(tail_losses)

VaR_95
cVaR_95
```

Investigation / Activity: Writing a cVaR Function

★★ Intermediate

Task. Write a reusable function that calculates cVaR.

R Code:

```
calculate_cVaR <- function(loss_values, alpha) {
  VaR_value <- calculate_VaR(loss_values, alpha)
  tail_losses <- loss_values[loss_values >= VaR_value]
  mean(tail_losses)
}

calculate_cVaR(losses, 0.90)
calculate_cVaR(losses, 0.95)
calculate_cVaR(losses, 0.99)
```

Reflection Question: VaR Versus cVaR

Why might a risk manager prefer cVaR when they care about what happens during very bad scenarios?

F. Parametric Normal VaR

Key Definition / Result

A parametric method assumes a probability model for the data. If losses are assumed to be approximately Normal:

$$L \sim N(\mu, \sigma),$$

then the α -level VaR is:

$$\text{VaR}_\alpha(L) = \mu + \sigma z_\alpha,$$

where

$$z_\alpha = \Phi^{-1}(\alpha).$$

In R:

```
qnorm(alpha, mean = mu, sd = sigma).
```

Worked Example: Normal VaR from Estimated Mean and Standard Deviation

★★

Intermediate

Question. Estimate the mean and standard deviation of the simulated loss values. Then calculate the 95% Normal VaR and compare it with the empirical VaR.

R Code:

```
mu <- mean(losses)
sigma <- sd(losses)

normal_VaR_95 <- qnorm(0.95, mean = mu, sd = sigma)
empirical_VaR_95 <- quantile(losses, 0.95)

normal_VaR_95
empirical_VaR_95
normal_VaR_95 - empirical_VaR_95
```

Common Mistake

Do not apply the Normal method blindly. If the loss distribution is discrete, skewed or strongly affected by rare events, a Normal approximation may misrepresent tail risk. The method is part of the answer. A VaR number without knowing whether it came from empirical data, a Normal model or simulation is incomplete.

G. Empirical, Parametric and Monte Carlo Methods

Key Definition / Result

There are three common approaches for measuring VaR:

Method	Main idea	Main limitation
Empirical method	Use sample quantiles from observed or simulated data	The sample may not represent the future
Parametric method	Assume a distribution such as Normal	Model may be wrong
Monte Carlo simulation	Simulate many possible outcomes from a chosen model	Depends on simulation assumptions

Investigation / Activity: Comparing Three VaR Methods *** Advanced

Task. Compare empirical VaR, Normal VaR and Monte Carlo VaR for the same loss dataset.

R Code:

```
set.seed(2026)
alpha <- 0.95

# Empirical method
empirical_VaR <- quantile(losses, alpha)

# Parametric Normal method
mu <- mean(losses)
sigma <- sd(losses)
normal_VaR <- qnorm(alpha, mean = mu, sd = sigma)

# Monte Carlo method using the Normal assumption
monte_carlo_losses <- rnorm(100000, mean = mu, sd = sigma)
monte_carlo_VaR <- quantile(monte_carlo_losses, alpha)

comparison_table <- data.frame(
  Method = c("Empirical", "Parametric Normal", "Monte Carlo Normal"),
  VaR_95 = c(empirical_VaR, normal_VaR, monte_carlo_VaR)
)
comparison_table
```

Reflection Question: Comparing Assumptions

If the empirical method, Normal method and Monte Carlo method give different answers, does this mean one of them must be wrong? Or does it mean they are asking the risk question under different assumptions?

H. VaR on Portfolios: Combining Risks, Weights and Diversification

Synergy & Comparison

From one asset to many assets: So far, VaR has been calculated for one vector of losses. In real risk management, however, a portfolio usually contains more than one asset, instrument or source of uncertainty. The risk of a portfolio is not simply the risk of Asset 1 plus the risk of Asset 2. It also depends on the portfolio weights and on whether the assets tend to move together.

Key Definition / Result

Suppose a portfolio contains two assets with returns R_1 and R_2 . If the portfolio weights are w_1 and w_2 , then the portfolio return is

$$R_P = w_1 R_1 + w_2 R_2.$$

If we define loss as the negative of return, then

$$L_P = -R_P.$$

For a portfolio loss variable L_P , the α -level portfolio VaR is

$$\text{VaR}_\alpha(L_P) = q_\alpha(L_P).$$

The important new idea is that portfolio risk depends not only on the individual risks of the assets, but also on how the assets move together.

Suggested AI Prompt: Explaining Portfolio VaR**★★ Intermediate**

“Explain portfolio VaR to gifted secondary school students. Start from the idea of combining two risky assets with weights. Explain why diversification can reduce risk, and why correlation matters. Use simple examples before showing R code.”

Worked Example: A Two-Asset Portfolio from Normal Returns

★★

Intermediate

Question. Suppose we simulate daily returns for two assets. Asset 1 has lower volatility, while Asset 2 has higher volatility. Construct a portfolio with 60% invested in Asset 1 and 40% invested in Asset 2. Then calculate the 95% VaR of the portfolio loss.

R Code:

```
set.seed(2026)
sample_size <- 1000

asset_1_returns <- rnorm(sample_size, mean = 0.0005, sd = 0.01)
asset_2_returns <- rnorm(sample_size, mean = 0.0008, sd = 0.02)

weight_1 <- 0.60
weight_2 <- 0.40

portfolio_returns <- weight_1 * asset_1_returns +
  weight_2 * asset_2_returns

portfolio_losses <- -portfolio_returns
portfolio_VaR_95 <- quantile(portfolio_losses, 0.95)

portfolio_VaR_95
```

Reflection Question: Weighted Average Does Not Mean Simple Risk Average

The portfolio return is a weighted average of asset returns. Does this mean that portfolio risk is always the same weighted average of individual asset risks? Why or why not?

Key Definition / Result

Diversification intuition. If two assets do not always move in the same direction, combining them may reduce portfolio risk. When one asset performs poorly, another asset may perform better. This is the basic intuition behind diversification.

However, diversification is weaker when assets are highly positively correlated. If many assets fall together during a crisis, the portfolio may still suffer a large loss.

Investigation / Activity: Comparing Individual VaR and Portfolio VaR**

Intermediate

Task. Calculate the 95% VaR of Asset 1, Asset 2 and the combined portfolio. Compare the results.

R Code:

```
asset_1_losses <- -asset_1_returns
asset_2_losses <- -asset_2_returns

asset_1_VaR_95 <- quantile(asset_1_losses, 0.95)
asset_2_VaR_95 <- quantile(asset_2_losses, 0.95)

weighted_average_VaR <- weight_1 * asset_1_VaR_95 +
  weight_2 * asset_2_VaR_95

VaR_comparison <- data.frame(
  Position = c("Asset 1", "Asset 2", "60/40 Portfolio",
               "Weighted Average of Individual VaRs"),
  VaR_95 = c(asset_1_VaR_95, asset_2_VaR_95,
              portfolio_VaR_95, weighted_average_VaR)
)

VaR_comparison
```

Common Mistake

Do not calculate portfolio VaR by simply adding the VaRs of individual assets unless the assumptions justify such a conservative calculation. VaR is a property of the whole portfolio loss distribution, not merely a list of separate asset VaRs.

Key Definition / Result

A simple parametric formula. If two asset returns have standard deviations σ_1 and σ_2 , and correlation ρ , then the portfolio variance is

$$\sigma_P^2 = w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \rho \sigma_1 \sigma_2.$$

The portfolio standard deviation is $\sigma_P = \sqrt{\sigma_P^2}$.

If portfolio returns are approximately Normal with mean μ_P and standard deviation σ_P , then a lower-tail return quantile may be calculated using `qnorm()`, and it can be converted into a loss-based VaR.

Worked Example: Parametric Normal Portfolio VaR

*** Advanced

Question. Estimate the mean and standard deviation of the simulated portfolio returns. Then calculate a Normal approximation to the 95% portfolio VaR.

R Code:

```
portfolio_mean <- mean(portfolio_returns)
portfolio_sd <- sd(portfolio_returns)

# Lower 5% return quantile
portfolio_return_quantile_5 <- qnorm(
  0.05,
  mean = portfolio_mean,
  sd = portfolio_sd
)

# Loss-based 95% VaR
normal_portfolio_VaR_95 <- -portfolio_return_quantile_5
normal_portfolio_VaR_95
portfolio_VaR_95
normal_portfolio_VaR_95 - portfolio_VaR_95
```

Investigation / Activity: Seeing Correlation in Action

*** Advanced

Task. Simulate two correlated Normal return series. Change the correlation value and observe how the portfolio VaR changes.

R Code:

```
set.seed(2026)
sample_size <- 1000
rho <- 0.70

z_1 <- rnorm(sample_size)
z_2 <- rho * z_1 + sqrt(1 - rho^2) * rnorm(sample_size)
cor(z_1, z_2)

asset_1_returns_cor <- 0.0005 + 0.01 * z_1
asset_2_returns_cor <- 0.0008 + 0.02 * z_2
cor(asset_1_returns_cor, asset_2_returns_cor)

portfolio_returns_cor <- weight_1 * asset_1_returns_cor +
  weight_2 * asset_2_returns_cor
portfolio_losses_cor <- -portfolio_returns_cor

portfolio_VaR_95_cor <- quantile(portfolio_losses_cor, 0.95)
portfolio_VaR_95_cor
```


Investigation / Activity: Correlation Scenario Table

*** Advanced

Task. Compare portfolio VaR under several different correlation levels.

R Code:

```
set.seed(2026)
rho_values <- c(0.00, 0.30, 0.70, 0.90)
portfolio_VaR_by_rho <- rep(0, length(rho_values))

for (i in 1:length(rho_values)) {
  rho <- rho_values[i]
  z_1 <- rnorm(sample_size)
  z_2 <- rho * z_1 + sqrt(1 - rho^2) * rnorm(sample_size)

  asset_1_returns_temp <- 0.0005 + 0.01 * z_1
  asset_2_returns_temp <- 0.0008 + 0.02 * z_2

  portfolio_returns_temp <- weight_1 * asset_1_returns_temp +
    weight_2 * asset_2_returns_temp
  portfolio_losses_temp <- -portfolio_returns_temp

  portfolio_VaR_by_rho[i] <- quantile(portfolio_losses_temp, 0.95)
}

correlation_table <- data.frame(
  Correlation = rho_values,
  Portfolio_VaR_95 = portfolio_VaR_by_rho
)
correlation_table
```

Investigation / Activity: Portfolio cVaR

*** Advanced

Task. Calculate the 95% cVaR of the portfolio using the empirical loss distribution.

R Code:

```
portfolio_VaR_95 <- quantile(portfolio_losses, 0.95)
portfolio_tail_losses <- portfolio_losses[
  portfolio_losses >= portfolio_VaR_95
]
portfolio_cVaR_95 <- mean(portfolio_tail_losses)

portfolio_VaR_95
portfolio_cVaR_95
```

I. Tail-Risk Visualisation

Investigation / Activity: Drawing Losses with VaR Lines ** Intermediate

Task. Using the generic simulated `losses` vector from the Poisson claim example, draw a histogram and add 95% and 99% VaR lines.

R Code:

```
VaR_95 <- quantile(losses, 0.95)
VaR_99 <- quantile(losses, 0.99)

hist(
  losses,
  breaks = 20,
  main = "Poisson Claim Losses with VaR Thresholds",
  xlab = "Loss Magnitude",
  col = "lightblue",
  border = "white"
)
abline(v = VaR_95, col = "red", lwd = 2, lty = 2)
abline(v = VaR_99, col = "darkred", lwd = 2, lty = 3)

legend(
  "topright",
  legend = c("95% VaR", "99% VaR"),
  col = c("red", "darkred"),
  lwd = 2,
  lty = c(2, 3),
  bty = "n"
)
```

Investigation / Activity: Visualising a Rare Shock Model ** Intermediate

Task. Compare the ordinary Poisson loss model with the Bernoulli shock model using histograms and VaR values.

R Code:

```
ordinary_VaR_95 <- quantile(ordinary_losses, 0.95)
shock_model_VaR_95 <- quantile(total_losses, 0.95)

ordinary_VaR_95
shock_model_VaR_95

par(mfrow = c(1, 2))
hist(
```

```

ordinary_losses,
breaks = 20,
main = "Ordinary Poisson Losses",
xlab = "Loss",
col = "lightblue",
border = "white"
)
abline(v = ordinary_VaR_95, col = "red", lwd = 2, lty = 2)

hist(
  total_losses,
  breaks = 30,
  main = "Losses with Rare Shock",
  xlab = "Loss",
  col = "lightblue",
  border = "white"
)
abline(v = shock_model_VaR_95, col = "red", lwd = 2, lty = 2)
par(mfrow = c(1, 1))

```

Reflection Question: Interpreting the Tail

If the 99% VaR is much larger than the 95% VaR, what might this suggest about the shape of the loss distribution?

J. Consolidation Questions

Investigation / Activity: Review Set A: Programming and Loops ★ Core

1. What is the purpose of defining a function in R?
2. What is the difference between a simple `if` statement and a vectorised logical expression?
3. What does an indicator function do?
4. Why are for loops useful when studying simulation variability?

Investigation / Activity: Review Set B: Quantiles

★ Core

1. What is a 95% quantile?

2. What R command calculates the sample 95% quantile of a vector called `x`?
3. What is the difference between `quantile(x, 0.95)` and `qnorm(0.95, mean, sd)`?
4. If you are modelling the number of daily accidents, which theoretical quantile function should you use: `qnorm`, `qunif`, `qpois`, or `qbinom`? Why?
5. Why might sample quantiles change if we collect more data?
6. Why are high quantiles important in risk management?

Investigation / Activity: Review Set C: VaR**** Intermediate**

1. Define VaR for a loss random variable.
2. Explain why VaR is a quantile-based risk measure.
3. What does 95% VaR tell us?
4. What does 95% VaR not tell us?
5. Why must we distinguish between returns and losses when interpreting VaR?

Investigation / Activity: Review Set D: cVaR**** Intermediate**

1. Define cVaR for a loss random variable.
2. Why is cVaR usually at least as large as VaR for the same confidence level?
3. Write R code to calculate the 95% cVaR from a vector called `losses`.
4. Why may cVaR be more useful than VaR during severe market stress?
5. Why can discrete data create extra care when calculating empirical cVaR?

Investigation / Activity: Review Set E: Portfolio VaR***** Advanced**

1. How do portfolio weights affect portfolio returns?
2. Why is portfolio VaR not always equal to the weighted average of individual asset VaRs?
3. What is diversification?

4. Why does correlation matter in portfolio risk?
5. What may happen to portfolio risk if correlation becomes higher during a crisis?

Investigation / Activity: Review Set F: Methods of Risk Measurement***
Advanced

1. What is the empirical method?
2. What is the parametric Normal method?
3. What is the Monte Carlo method?
4. Why can different methods give different VaR estimates?
5. Why is a VaR number incomplete if we do not know the method used to calculate it?
6. In what situation might past or simulated data understate future risk?

Investigation / Activity: Challenge Questions

*** Challenge

1. Suppose two investments have the same 95% VaR but very different 95% cVaR. Which one may be more dangerous in extreme scenarios? Explain.
2. Why can a model with a poor tail assumption produce misleading VaR?
3. How would you explain the difference between randomness and model uncertainty?
4. If the data contain one extremely large loss, how may this affect VaR and cVaR differently?
5. If two assets become more strongly positively correlated during a crisis, what may happen to the diversification benefit?
6. In the Bernoulli shock model, why can a rare event still matter greatly for risk management?
7. Why might a Normal model be convenient but still risky when modelling losses?

End of Tutorial 8